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Estimating years of life lost due to premature mortality at regional level in France in 2017, using a probabilistic redistribution approach

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Abstract

Background Years of life lost (YLL) due to premature mortality is an important metric to assess the fatal impact of diseases. The main objectives of this study were to apply the four-step probabilistic method to redistribute ill-defined deaths (IDDs) and to quantify the premature mortality burden at regional level in France for 2017.

Methods We used the statistical database on medical causes of death derived from death certificate collection and coded by the Center for Epidemiology on Medical Causes of deaths (INSERM-CépiDc). First, we mapped the specific ICD-10 codes that define the underlying cause of death (CoD) to the Global Burden of Disease (GBD) cause list. Second, identified IDDs were redistributed to specific ICD-10 codes. A four-step probabilistic redistribution developed for the Belgium Burden of Disease (BeBOD) study was adopted to fit the French context: redistribution using predefined ICD codes, package redistribution using multiple causes of death data, internal redistribution, and redistribution to all causes. Finally, Standard Expected Years of Life Lost (SEYLL) and age-standardized SEYLL rates (ASYR) were calculated at regional level, using the GBD 2019 reference life table.

Results In France, 36% of all deaths were IDDs in 2017. The majority was redistributed using predefined ICD codes (14%), followed by the package redistribution using multiple causes of death data (11%), all-cause redistribution (11%) and internal redistribution (< 1%). The total number of SEYLL was 9.6 million for all causes, (4.1 million in females [43%] and 5.5 million in males [57%]). Tracheal, bronchus, and lung cancer ranked first (10%), followed by ischemic heart disease (7%), and Alzheimer's disease and other dementias (6%) in terms of SEYLL. For all causes, we observed the lowest ASYRs in Corse for females (8970 per 100 000) and in Ile-de-France for males (16 109 per 100 000).

Conclusions We quantified the full mortality burden for the first time in France at regional level, based on a new probabilistic redistribution method developed by researchers from Sciensano before COVID-19 pandemic. These estimates are important for future investigations on the contribution of social inequalities and risk factors to all-cause mortality in France with a focus on regional differences.

Keywords Cause of death, Mortality, Years of life lost, YLL, Redistribution, Ill-defined deaths, Garbage Code, Life expectancy, France

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Background

Burden of disease (BoD) studies are an increasingly popular means to assess the population health impact of different health conditions, comprehensively and comparably according to a standardized concept [1, 2]. Disability-Adjusted Life Years (DALYs) are one of the most commonly used health gap summary measures in the field of public health and have become a key metric for quantifying the burden of disease. A health gap measure quantifies the difference between the observed population health state and a hypothetical one where everybody reaches old age free of disease or injury [3]. DALYs summarize the years of life lost due to premature mortality (YLL) and the years of life lived with disability (YLD) into a population-based composite indicator [4]. This metric allows direct comparisons of various diseases and injuries and informs discussions related to the prioritization of preventive and interventional measures, usually based on demographic-specific needs.

Causes of death (CoD) statistics are an important source of information for epidemiological research and can be used in public health policy decisions to develop interventions and prevention policies [5]. Official mortality statistics are routinely based on the underlying CoD, which is the disease or injury that initiated the chain of events leading to death [6]. However, not all coded underlying causes of death are appropriate to estimate the BoD [5]. Some ICD (international statistical classification of diseases and related health problems 10th revision) codes do not contain enough specific information surrounding the death (e.g., ICD-10 A49 “Bacterial infection of unspecified site”), while other codes represent inaccurate causal sequences (e.g., ICD-10 I50 “Heart failure” as the underlying CoD). Some CoDs are not regarded as sufficiently specific causes of death for public health purposes. These CoDs are usually referred to as “ill-defined deaths” (IDDs) or “garbage codes” [5]. Their proportion varies considerably between countries from 7% (Finland) to 66% (Egypt) [7]. However, the quality of certification can be affected by: (i) lack of information in identifying the underlying CoD; (ii) lack of importance attributed to death certification; (iii) mistakes made during data generation; and (iv) lack of training of physicians in this specific task [8]. The presence of IDDs is a major challenge for the calculation of YLL (years of life lost due to premature mortality). Therefore, to estimate the BoD accurately, the IDDs need to be redistributed to specific categories of causes of death, represented by ‘target codes’ [9]. Redistribution is the process of reallocating IDDs to plausible underlying causes (i.e., target causes) as defined in the Global Burden of Disease (GBD) study [10]. Currently, there are a number of methodological approaches using multiple cause of death data (MCOD)

to redistribute IDDs [11–15]. Each method has its advantages. Proportional redistribution allocates IDDs to specific causes based on the distribution of known causes of death in a reference population. It is easy to implement, ensures the consistency with observed cause of death patterns and it works well in stable populations [14, 16]. Probabilistic redistribution, on the other hand, assigns IDDs based on known probabilities derived from various covariates and contextual data. The main advantages of this approach include greater precision, by taking into account factors such as age and sex, it reduces bias from the assumption that IDDs follow the same distribution as specific causes and it can be tailored to national datasets, offering flexibility for context specific adaption [10, 15].

Along with the underlying CoD, death certificates often contain the sequence of events or diseases that resulted in death, as well as other causes contributing to death (i.e., contributive or associated causes of death). This additional information on the death certificate that accompanies or (in) directly causes the death, can be used to validate the registered underlying CoD or complement the information in the coded underlying CoD [15].

Within the framework of the GBD study, one of the most well-known methods was developed by Naghavi and colleagues, integrating fractional reassignment of death due to multiple causes, proportional reassignment, regression models, and redistribution based on fixed proportions [10]. The GBD algorithm was last revised in GBD 2013 with some minor updates in GBD 2019 and GBD 2020, and can be used to derive country-specific estimates [9, 17]. Generic redistribution techniques enable international comparisons; nevertheless, they may produce biased estimates due to differential patterns in IDDs observed between and within countries [7, 8]. GBD estimates seem to be useful to compare population health between countries but possibly less appropriate for comparisons within countries. However, many national decision-making processes count on those internal comparisons over time and between groups. To date, there is a number of literature on national BoD assessments [e.g., [14, 15, 18–20]]. Recently, the Belgian national burden of disease study (BeBOD) developed a probabilistic redistribution approach using multiple causes of death data to generate DALY estimate using local data sources and evidence [15]. The CoD of death statistics coded using world health organization (WHO) ICD-10 including underlying and multiple causes of death, have similarities in data structure in France and Belgium. Our main hypotheses were: *first*, that the four-step redistribution method from the BeBoD study is applicable to the French multiple causes of death dataset without significant adaptation; *second*, that the redistribution of IDDs using multiple CoD data leads to more accurate and regionally

consistent CoD profiles; and *third*, that there were regional differences in the premature mortality burden in France in 2017. Therefore, for this study, we applied the redistribution approach developed by Sciensano (i.e., in European context) to gain an in-depth understanding of this process including various technical aspects and to learn new methodologies. We adapted this method to the French mortality database to generate cause-specific YLL mortality estimates based on local data sources, before COVID-19 pandemic. This study has two main objectives: *firstly*, to apply the four-step method used in BeBOD to redistribute IDD, using multiple causes of death data; and *secondly*, to provide a quantification of the premature mortality burden at regional level in France for 2017.

Methods

Causes of death data

We used the French cause-of-death statistical database for the year 2017 including all causes of deaths occurring in France. Each year, INSERM-CépiDc produces the national statistics of medical causes of death in France based on death certificates collection and coding [21]. Medical parts of death certificates are received by INSERM-CépiDc and are coded according to the rules of the 10th revision of the international statistical classification of diseases and related health problems on 3–4 digits of the WHO [6]. In France, death certificates can be completed by physicians electronically or manually [21]. In 2017, the deployment of e-certificates was around 12% with both regional and place of death types of discrepancies [22], increasing to approximately 32% of total deaths that occurred in France in 2021 [23, 24]. Electronic death registration system can enhance the accuracy and timeliness of cause-of-death data, potentially reducing the incidence of ill-defined causes of death. Studies have indicated that electronic death certificates may improve data quality by minimizing errors and omissions associated with manual, paper-based systems [25]. The medical part of death certificates follows the WHO international standard as stated in the ICD-10 guides [26] having two parts: *Part I* includes four lines that allow the certified physician to describe the causal chain of diseases that directly led to death, from the immediate CoD reported on the first line to the initial CoD mentioned on the last line filled. *Part II* includes other causes that allow notification of morbid states that may have contributed to the CoD. In 2017, almost 50% of death certificates were coded by an automated, rule-based expert system, the IRIS software, which is maintained internationally [27], and 50% were manually coded by the coding experts of INSERM-CépiDc using IRIS interactively [28]. All causes of death appearing in the certificates were coded and the

primary or main or underlying cause was determined following ICD-10 rules.

The causes-of-death database is part of the French National Health Data System (i.e., SNDS [*Système National de Données Santé*]) that is used for public health surveillance in France [29, 30]. We selected 2017 as the reference year to calculate the YLL, as it was the most recent year available at the time of analysis, with verified data quality and full population coverage at the time of study. For this study, we included all those individuals who died in 2017 with residence status in France including 13 metropolitan and five overseas regions (i.e., Guadeloupe, Guyane, La Réunion, Martinique, and Mayotte).

Mapping of WHO ICD-10 codes to GBD causes

We mapped every ICD-10 code of the underlying CoD of each individual to the GBD 2019 cause list [31]. The GBD 2019 classification of diseases and injuries was adopted to enhance the comparability of data for causes of death with other countries and it has been designed for burden of disease analysis [32]. The GBD classification includes three groups with 4 levels. In French mortality data, the final cause map included 140 causes of death (level 3), grouped into 21 disease groups (level 2), each classified into one of the three GBD groups (level 1), i.e., group 1: non-communicable diseases (NCD), group 2: communicable, maternal, neonatal and nutritional diseases, and group 3: injuries. There were some causes of death on level 3, which were further subdivided into two or more level 4 causes. For instance, diabetes mellitus (level 3) was subdivided into type 1 diabetes (level 4) and type 2 diabetes (level 4). Following the Belgium approach, we mapped CoDs at level 4, but the results were presented at level 3. In the mapping process, certain ICD-10 codes could not be matched with a specific level 4 GBD cause such as asthma and chronic obstructive pulmonary disease. For these cases, the redistribution process of IDD was performed by taking into account the level 3.

Redistribution of ill-defined deaths (IDDs)

We applied the Belgian four-step probabilistic approach for the redistribution of IDD, adapted to the French mortality databases [15]. We followed the four-step redistribution approach: 1. Predefined ICD-based redistribution; 2. Package redistribution based on multiple causes of death (MCOD); 3. Internal redistribution; and 4. All-cause redistribution. The target distributions were stratified by six age groups (0–4, 5–14, 15–44, 45–64, 65–84, 85+) and sex to have more precision in the redistribution process. The detailed method has been described elsewhere [15]. Here is a brief description of the four-step redistribution applied to French mortality dataset: 1. Predefined ICD-based

redistribution: In this step, the selected underlying CoDs of IDD were proportionally redistributed to predefined ICD-10 target codes « ICD-based redistribution». The predefined ICD-10 target codes have been adopted based on the methodology applied previously used in Belgium and Scotland [15, 20]. For example, non-diabetic hypoglycaemic coma (ICD-10: E15) is an IDD and is redistributed to two target codes: type 1 diabetes (ICD-10: E10) and type 2 diabetes (ICD-10: E11), according to the occurrence of each disease as underlying CoD. The target distributions were defined based on the deaths that occurred in 2017. This first step of redistribution process was only applicable to IDDs with predefined ICD-10 codes providing clear information on the possible underlying CoD, and thus allow defining target codes a priori [15].

2. Package redistribution based on MCODE: In the second step, we used the French multiple causes of death data to define targets and redistribute selected remaining IDDs that were coded as an uninformative underlying CoD code (“package redistribution”). For example, the assigned underlying CoD was “Acute respiratory distress syndrome” (ICD-10 J80), which is an IDD and there are no predefined target codes available at step 1. For these cases, we defined packages, i.e., containing sets of IDDs that were similar and considered to have a similar redistribution target. In this example, the package created is called “Acute respiratory failure” and consists of the following ill-defined ICD-10 codes: J800, J96, J960, J969, J981, and J982. In this step, an IDD was assigned to a package based on its underlying CoD and we selected the MCODEs including some/several specific codes that were used as targets. These targets were the underlying CoD of the well-defined deaths that have one of the IDDs as MCODE and we randomly selected one of the specific codes as targets. The redistribution was performed proportionally to the observed target codes, stratified by age group and sex, as the target was defined in function of the multiple causes of death data of the year 2017, corresponding to each IDD including in the package of IDDs, already defined by the GBD.

3. Internal redistribution: In the third step, we performed an internal redistribution of IDDs, to fully use the available multiple causes of death data. In this approach, the remaining IDDs were randomly assigned to a specific GBD cause that was mentioned as contributive or associated CoD on the death certificate of the deceased individual.

4. All-cause redistribution: In the final step, all remaining IDDs, including all codes of ICD-chapter R, “Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified”, were assigned as IDDs. These were proportionally redistributed

overall specific and already redistributed causes having occurred during 2017, stratified by age group and sex.

We excluded the GBD causes (level 3) with less than five occurrences in 2017 as possible targets. These rare causes (e.g., Chagas disease [B57], Leishmaniasis [B55]) included very specific diseases and would therefore be rarely missed as underlying CoD. This 4-step probabilistic redistribution approach was performed using 30 iterations to capture the uncertainty in the results. The number of iterations was limited to 30 due to the limited computational capacity of the SNDS servers. Each iteration was a complete run of the four-step process and resulted in a completely imputed CoD dataset. For each of the summary statistics (as described below), 30 random estimates were thus generated. The means and 95% uncertainty intervals of these 30 random estimates were calculated, the latter defined as the 2.5th and 97.5th percentile of the random estimates.

Quantification of premature mortality burden of disease in France

We calculated the number of deaths and premature mortality burden in France using the Standard Expected Years of Life Lost (SEYLL), which is a well-established measure for the burden of premature mortality [33]. SEYLL is defined as the total number of years that persons in a population would have lived if they had not died prematurely due to a disease or injury, against an aspirational life expectancy based on the lowest age-specific mortality rates observed across time and geographical locations. The deaths at each specific age can be weighted by the expected number of years lost at each age. Based on the recommendations made in previous studies, the GBD 2019 reference life table was used for calculating the years of life lost for each age group [34–36].

The number of deaths and SEYLL as absolute numbers, crude rates (per 100,000 inhabitants), and age-standardized rates were presented based on five-year age groups. We calculated age-standardised rates by applying the age structure of the 2013 European Standard Population [37] to the observed age-specific rates in France, which were calculated using French population [38] to facilitate like-for-like comparisons across regions.

The choropleths maps with a scale of 12 colours varying from dark navy blue to yellow were produced for males and females, respectively. The cut-off colour represents the reference value of SEYLL, which is average level of SEYLL for both females and males allowing to easily compare the maps. We applied 1.1 step between the different categories of the scales, which means that each next threshold of a colour level is 1.1 times higher than the previous one. Therefore, the larger disparities between the regions, more colour will be presented. The

legend of the maps displays the boundaries of each category of the scale. The midpoint of each category of the scale is calculated as the average rate $\times (1.1)^{\text{rank of the scale}}$, starting from the average category of the scale [39].

All analyses were performed in R 4.1.0. The R scripts and inputs for the mapping and redistributions are available on request. This study is reported in adherence with the STROBE statement [40].

Ethical approval

The study was based on anonymous data from the national mandatory mortality database (INSERM-CépiDc). In accordance with French law, Santé publique France (SpF) has been granted specific access to this national mortality database in order to carry out its mission of health monitoring and health crisis response (article L. 1413–7 code de la santé publique). This study does not require the ethics approval and consent to participate.

Results

Description of ill-defined deaths

In the French cause-of-death database, IDD accounted for 36% of all deaths in 2017. Of these IDD-related deaths, 73% were recorded without R-codes, while 27% included R-codes. Within this 27% of R-codes, 12% were specifically coded as R99, indicating either no death certificate or insufficient information on the certificate (Supplementary material 1: Additional file 1). The level of IDD was stable for the period (2013–2017), ranging between 35–36%. IDDs were slightly more common in females (38%) than in men (34%). The percentage of IDDs were highest among age groups of 15–29 and 90 years and above, compared to other age groups. The percentage of IDDs was higher among overseas departments, especially in Mayotte (i.e., 55%), compared to the metropolitan regions in France. The percentage of IDDs was highest for death that occurred on public road/in the street (i.e., 55%) followed by deaths occurring at home (i.e., 46%) (Supplementary material 1: Additional file 2).

We also compared the regional distribution of IDDs (i.e., the percentage of IDDs relative to the total number of deaths in each region) between the two countries and found that the proportions were generally higher in France than in Belgium. For instance, the border regions of Grand Est and Hauts-de-France in France had higher percentages compared to the Wallonia region in Belgium (34% and 36% vs. 31%, respectively) (Fig. 1).

Redistribution of ill-defined deaths

Among 36% of the total deaths, which were IDDs, the majority were redistributed using predefined ICD codes in step 1 (14% of total deaths), followed by the package

redistribution in step 2 (11%) and all-cause redistribution step 4 (11%). Only a small number of IDDs were redistributed using the internal redistribution approach in step 3 (< 1%). Figure 2 provides an overview of the redistribution method with the percentage of redistributed deaths per step in 2017 (Supplementary material 2: Fig. S1).

In the French cause-of-death database, the most common IDDs as underlying causes were I50.9 (Heart failure, unspecified = 7%), followed by I64 (stroke, not specified as haemorrhage or infarction = 5%), J18.9 (Pneumonia, unspecified = 5%) and R99 (Ill-defined and unknown cause of mortality = 12%).

After the redistribution process, the causes with the largest number of redistributed IDDs using predefined ICD target codes (step 1) were lower respiratory infections, stroke, falls and trachea, bronchus and lung cancer (Fig. 3). Ischemic heart disease and Alzheimer's disease and other dementia has the largest share of package redistribution (step 2). Alzheimer's disease and other dementia, ischemic heart disease, stroke and trachea, bronchus and lung cancer had the largest share of all-cause redistribution (step 4).

After the redistribution, we compared the relative distribution of cause-specific allocation (i.e., specific causes of death defined prior to the redistribution process) and the deaths reassigned during the redistribution process (i.e., redistribution allocation) to the total number of causes of death. For example, other neurological disorders (86% cause-specific allocation + 14% from redistribution), aortic aneurysm (81% cause-specific allocation + 19% from redistribution) have a higher cause-specific allocation to the total number of relative deaths than from redistribution (Supplementary material 2: Fig. S2). On the other hand, lower respiratory infection (18% cause-specific allocation + 82% from redistribution) and other cardiovascular and circulatory diseases (23% cause-specific allocation + 77% from redistribution) have a higher contribution from redistribution to the total number of relative deaths than the cause-specific allocation (Supplementary material 2: Fig. S2). We also compared the most frequent (top 30) causes of death before and after the redistribution process (Supplementary material 2: Fig. S3). Lower respiratory infections, initially ranked 22nd, moved up to 6th place after redistribution. Similarly, other cardiovascular and circulatory diseases, originally ranked 26th, shifted to 8th after redistribution (Supplementary material 2: Fig. S3).

The impact of the redistribution of ill-defined deaths (IDD) was described in « Supplementary Material 2: Table S1 », by comparing the age-standardized mortality rates (ASMRs) between the two countries.

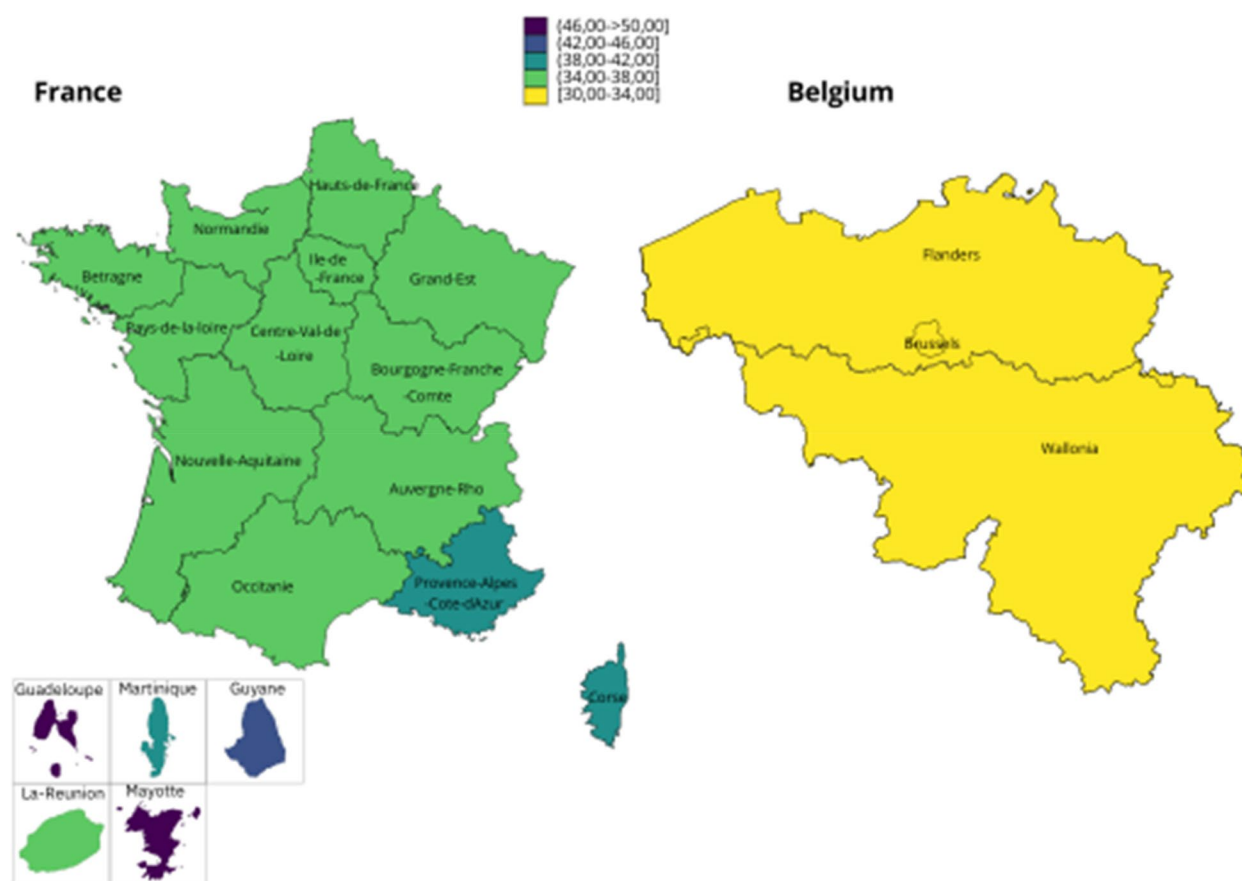


Fig. 1 Percentage of IDD's relative to the total number of deaths in each region of France and Belgium, 2017

Premature mortality burden of disease in France in 2017

After the redistribution process, we quantified the premature mortality burden of disease in France for 2017 in terms of number of deaths and SEYLL.

Number of deaths by GBD cause

The total number of deaths among French inhabitants was 601 643 in 2017 (age-standardized mortality rate 841 per 100 000 population) including 13 metropolitan regions and five overseas regions (Guadeloupe, Guyane, La Réunion, Martinique, Mayotte). Female and male deaths were almost equally distributed. Most deaths occurred in elderly people, with 78% of female deaths and 58% of male deaths at the age of 75 and above in 2017 (Fig. 4).

Before and after the redistribution process, the contribution of non-communicable diseases (level 1) remained the most dominant cause of mortality increasing from 58 to 85%, whereas the injuries (from 4 to 8%) and communicable diseases (from 2 to 7%) contributed comparatively less to the all-cause mortality in 2017 (Fig. 4) (Supplementary material 2: Fig. S4).

The distribution of these CoDs varied markedly across different age groups and both sexes when analyzing the subcategories (level 2) (Fig. 5). For example, neoplasms were the most dominant CoD, with a higher among males than females (39% vs 30%), followed by cardiovascular diseases, which were more common among females than males (24% vs 21%) (Fig. 5). In both sexes, neoplasms contributed significantly to mortality in children under 15 years of age, but their contribution grew markedly with increasing age, and was highest in the age group of 50–74 years age group. Cardiovascular diseases showed a steady increase from the 60–64 years age group until 95 years and older. Neurological disorders begin to rise in the 70–74 years age group and were the highest among 90 years and older, with a higher among females than males (15% vs 9%). Self-harm and interpersonal violence, on the other hand contributed the most to the age groups of 20–39 years and 15–29 years, with a substantially higher proportion among males than females (28% vs 1%), respectively (Fig. 5).

The level 3 GBD causes that included more specific disease and injury level, Alzheimer's disease and other

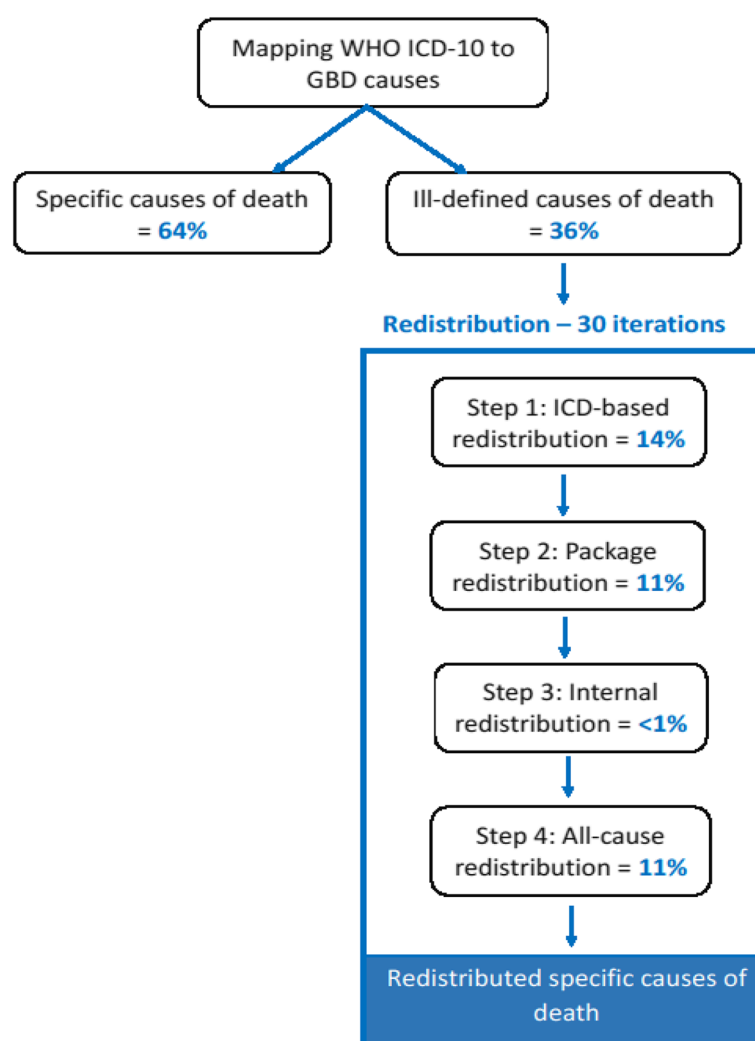


Fig. 2 The redistribution method with the percentage of redistributed deaths per step in French mortality database, 2017

dementia was the most dominant CoD (9%), followed by ischemic heart disease (8%), lung cancer (7%), and stroke (7%).

The age-standardized mortality rates (ASMRs) were 920 per 100 000 in males and 879 per 100 000 in females. The ASMRs for level 3 causes were substantially higher for males than females for lung cancer (106 per 100 000 vs 35 per 100 000) and ischemic heart disease (103 per 100 000 vs 42 per 100 000). For Alzheimer's disease and other dementia, these estimates were slightly higher among females than males (75 per 100 000 vs 68 per 100 000) (Supplementary material 2: Fig. S5). Regarding the regional variations, there were six regions (Nouvelle-Aquitaine, Bourgogne-Franche-Comte, Bretagne, Centre-Val-de-Loire, Provence-Alpes-Côte-d'Azur and Occitanie) where these ASMRs were higher than 1000 per 100 000 inhabitants.

Standard expected years of life lost (SEYLL)

The total crude number of SEYLL was 9.6 million for all causes, (4.1 million in females [43%] and 5.5 million in males [57%]). Among the major disease and injury group of level 1, non-communicable diseases were the dominant cause of SEYLL (84%), followed by injuries (12%) and communicable diseases (4%). At the subcategories of level 2, neoplasms were the leading cause of SEYLL with a contribution of 41%, followed by cardiovascular diseases (18%) and neurological disorders (9%). Level 3 focuses on more specific disease and injury level, including tracheal, bronchus, and lung cancer ranked first (10%), followed by ischemic heart disease (7%) and Alzheimer's disease and other dementias (6%).

The age-standardized SEYLL rates (ASYRs) were 14 264 per 100 000 inhabitants, higher among males (18 900 per 100 000) than females (10 389 per 100 000). For all causes, the regional ASYRs were highest for Mayotte for

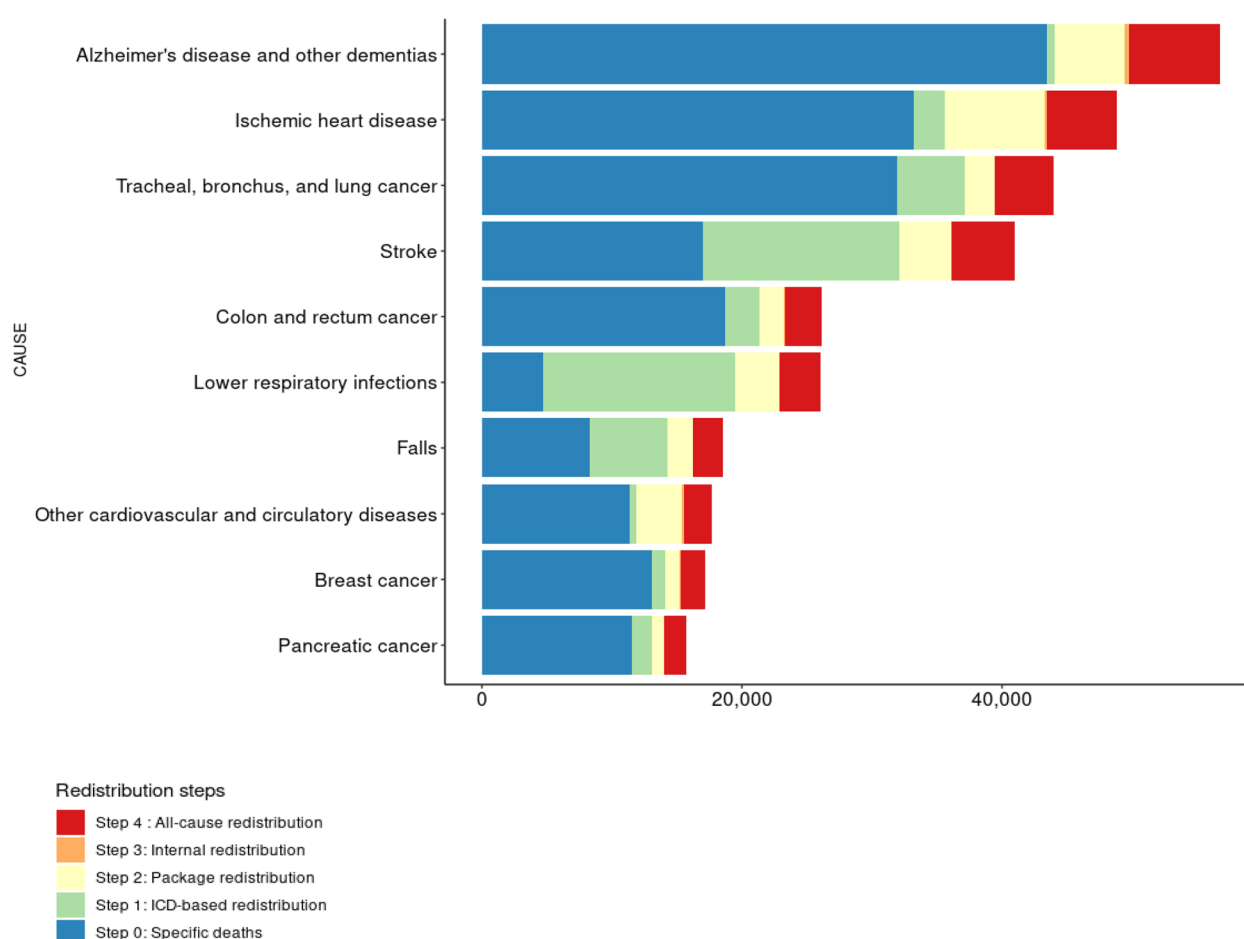


Fig. 3 Number of deaths, level 3 GBD cause, for the top 10 causes of death by the contribution of different redistribution steps, France, 2017

both sexes (females: 25 156 per 100 000, males: 25 905 per 100 000), followed by Guyane (females: 14 253 per 100 000, males: 23 417 per 100 000) (Fig. 6). We observed the lowest ASYRs in Corse for females (8970 per 100 000) and in Ile-de-France for males (16 109 per 100 000). While analyzing the leading causes of death by ASYRs (level 3) overall by sex (not shown), the estimates were higher for males than females for some diseases, such as lung cancer (2305 vs 816 per 100 000), ischemic heart disease (1617 vs 518 per 100 000) and stroke (866 vs 620 per 100 000); whereas Alzheimer's disease and other dementias were slightly higher among females than males (732 vs 715 per 100 000).

Figure 7 highlights the ASYRs of 15 leading causes of premature death by regions in Metropolitan France. The metropolitan regional trend of these causes of death follows the same patterns found in whole France. The lung cancer and ischemic heart disease were ranked first and second, respectively for all metropolitan regions (Fig. 7). One marked difference was self-harm, which was ranked third in Bretagne region and eighth in Ile-de-France

when compared with the sixth place in the French ranking (Fig. 7).

In overseas regions, for most of the causes higher ASYRs were found in Mayotte, followed by Guyane and La Réunion (Supplementary material 2: Fig. S6). The overseas regional trend of certain causes of death does not follow the same patterns found in whole France. For example, stroke was ranked first place in Mayotte compared with third in whole France, diabetes mellitus ranked second place in Mayotte whereas it ranked 12 in France.

Discussion

In the French cause-of-death database, 36% of all deaths were considered IDD, according to the GBD definitions and redistributed to specific CoD through four steps such as ICD-based redistribution (14%), package redistribution (11%), internal distribution (< 1%), and all-cause redistribution (11%). The leading causes of death were Alzheimer's disease and other dementias, ischemic heart disease and lung cancer, whereas lung cancer, ischemic

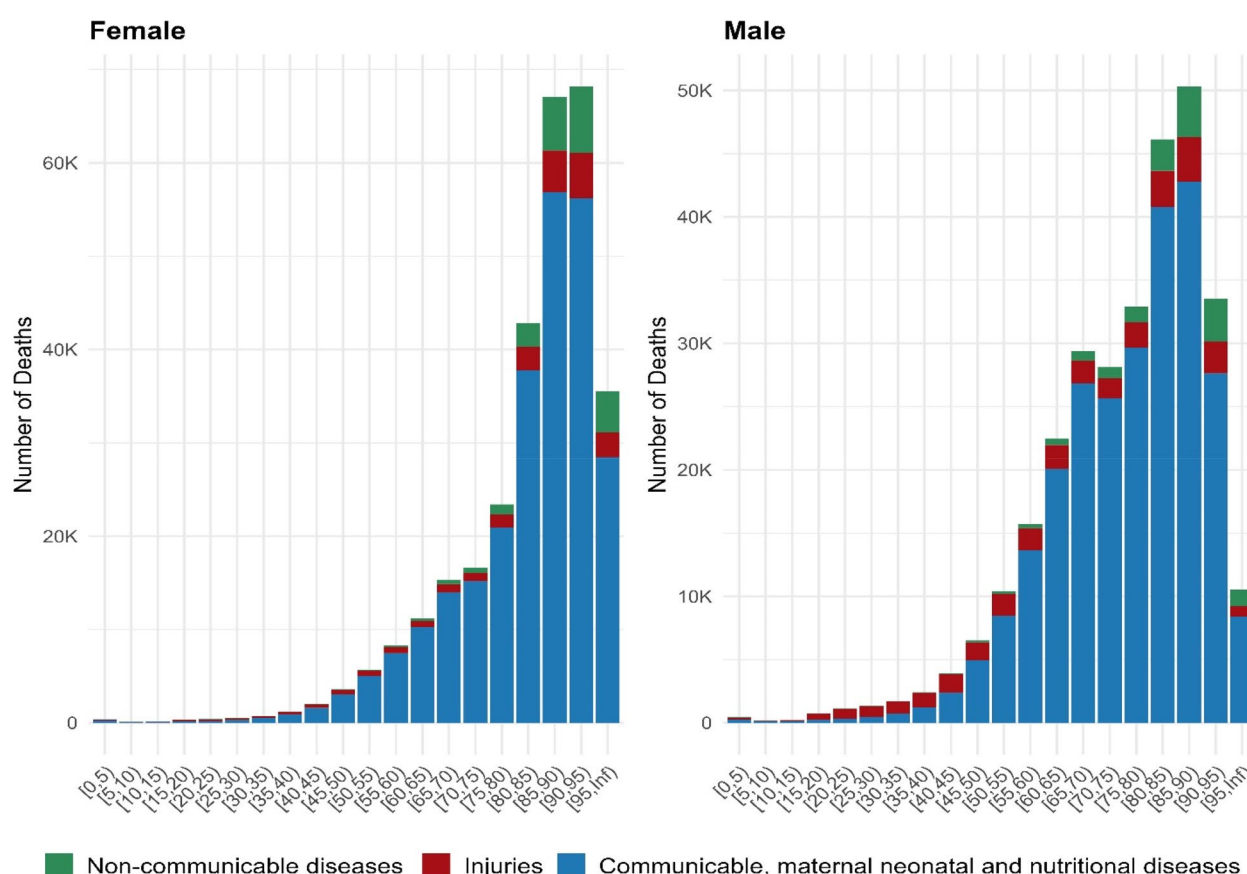


Fig. 4 Crude number of deaths, level 1 GBD cause by sex and age group after the redistribution of IDD in France, 2017

heart disease, stroke and Alzheimer's disease and other dementias were the main contributors to for premature mortality. The ASYRs were higher among males than females. We observed marked variation among metropolitan regions and overseas departments.

We observed the high proportion of IDD for the age groups 15–29 and 90 years and above. For the younger age group, it can be presumed that these deaths may be related to transport deaths occurred in the public road, suicide or drug use-related deaths. For deaths related to external death, when a medical or legal investigation is required, the medical part of the death certificate may not eventually be collected by the INSERM-CépiDc. The under-coverage rate is about 2% of all deaths at the national level, with regional discrepancies. Deaths not collected by INSERM-CépiDc are present in the cause-of-death database, due to a linkage with INSEE vital statistics, but with an ill-defined underlying CoD (R99) [23]. Moreover, in 2017, the template of death certificate did not cover a section on “manner of death”, in which the physician could specify whether the death was violent, natural, potentially related to suicide. This may have them under-report suicides and over-report ill-defined

external causes of death. A retrospective survey conducted in 2006 including certifying physicians and asking them to review some death certificates that ended coded as ill-defined external causes or as R99, and that may “hide” suicides concluded that suicide counts at that time was underestimated by 10% [41]. The high proportion of IDD can also potentially be explained by an ageing population as reflected in French mortality statistics with almost two-thirds of all deaths occurring in elderly people. Often deaths in old age are the results of a general deterioration of health with a complex multifactorial chain of events where no single cause can be attributed [15]. Furthermore, medical investigations are limited in the elderly population, therefore, precise diagnoses may be lacking [15]. Previous studies on redistribution of IDD, also mentioned the high share of redistributed deaths to old age [14, 42]. Therefore, among ageing societies, this underlines the importance of redistribution methods in mortality research and the need for other indicators than the underlying COD, to describe mortality of the elderly people more appropriately.

The total share of IDD in France was highest (i.e., 36%) as compared to other European countries such as

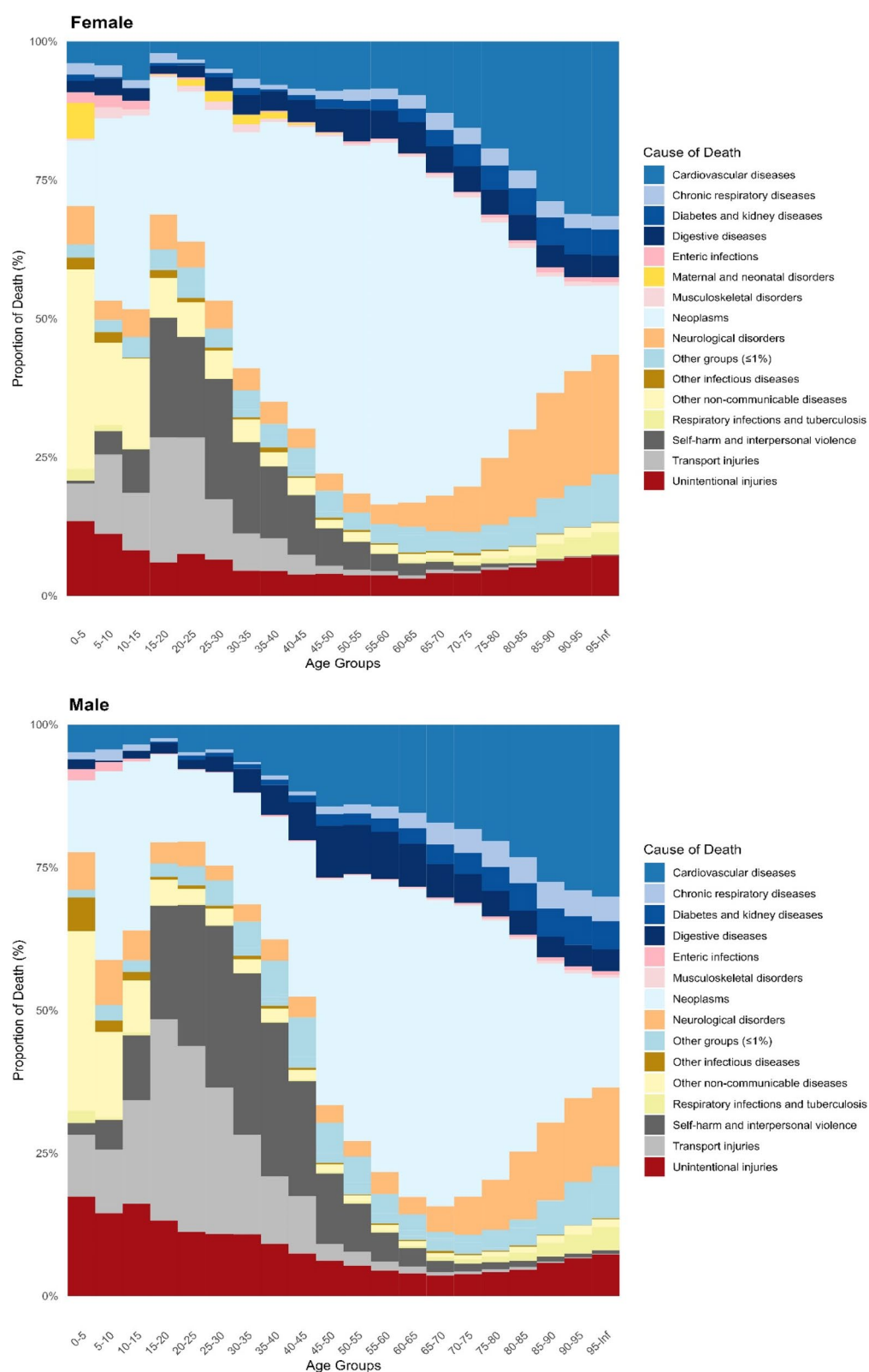


Fig. 5 Distribution of causes of death, level 2 GBD cause by age group and sex in France, 2017

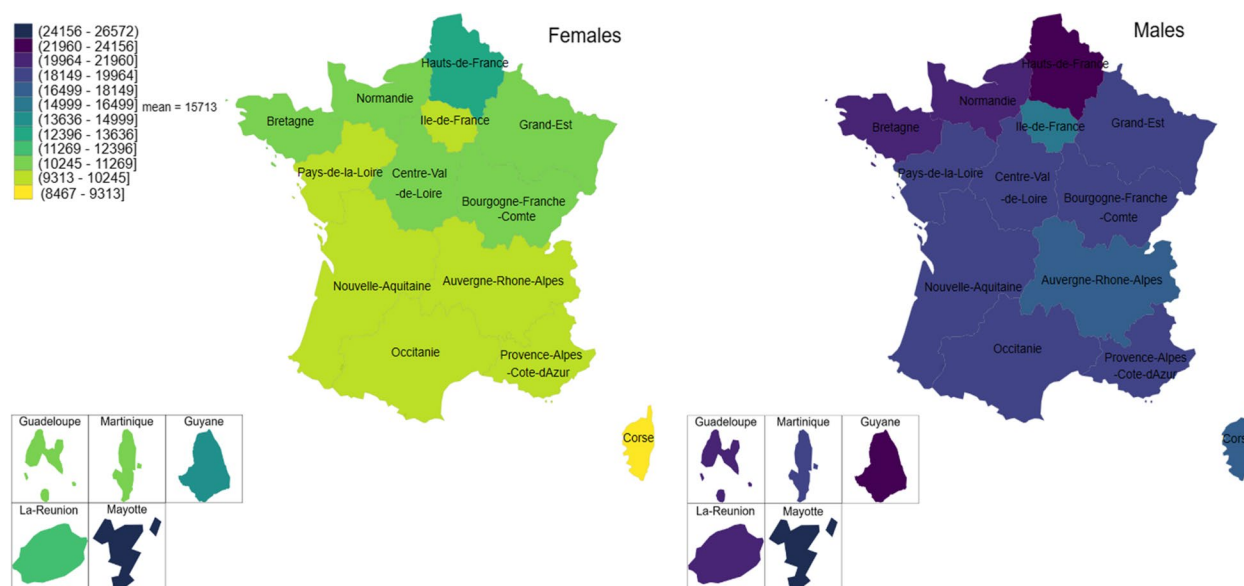


Fig. 6 Age-standardized years of life lost (ASYRs) (per 100 000 inhabitants), by sex and regions in France in 2017

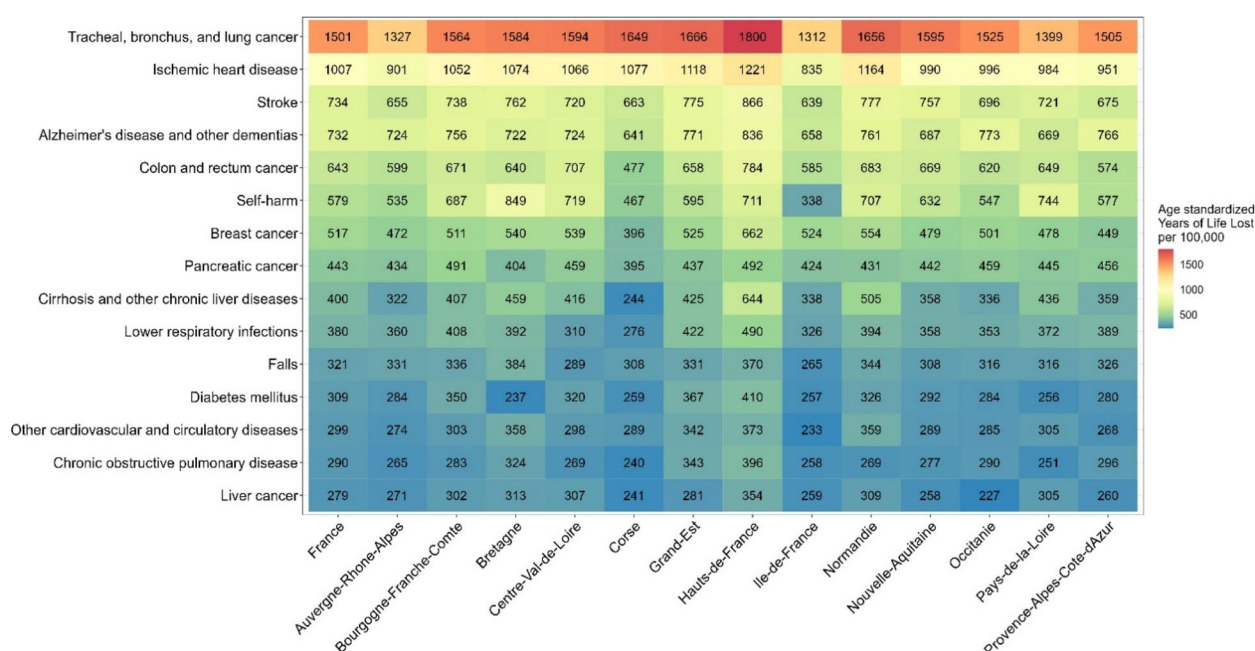


Fig. 7 15 Leading causes of premature death ranked by age-standardized years of life lost rate (ASYRs) (per 100 000 inhabitants), level 3 GBD cause, both sexes, Metropolitan France, 2017

Belgium (34%) [15], Germany (26%) [14] and Slovenia (20%) in 2017. The most common IDD as underlying causes were I50.9 (Heart failure, unspecified), followed by J18.9 (Pneumonia, unspecified) in Belgium mortality database [15], whereas in Germany, I50.9 (Heart failure, unspecified) and I64 (stroke, not specified as hemorrhage or infarction) were the most common IDDs [14]. These

results were aligned with the French cause-of-death database. As mentioned earlier, the most common IDDs as underlying causes were I50.9 (Heart failure, unspecified), followed by I64 (stroke, not specified as hemorrhage or infarction), J18.9 (Pneumonia, unspecified) and R99 (Ill-defined and unknown cause of mortality), which are aligned to international findings [7]. I64 and J18.9

are IDD at level 4 of GBD cause list and are specific to level 3 GBD causes of « lower respiratory infections» and « stroke». These IDDs were redistributed to the corresponding level 3 GBD causes and can explain the high share of redistributed deaths for lower respiratory infections and stroke. The high number of heart failure deaths defined as IDDs is commonly redistributed to ischemic heart disease of level 3 GBD causes. After redistribution of IDDs, the largest contribution to the French mortality burden was Alzheimer's disease and other dementia, ischemic heart disease and lung cancer.

We observed a high proportion of IDD in France that remained constant over 2013–2017. Furthermore, the regional differences for IDDs were more prominent between metropolitan regions and overseas departments. There are several reasons that could contribute to these differences, such as challenges in the death certification process, missing data ending up to undefined underlying CoD, lack of information or inadequate reporting, in case of medical investigation or if a person has several conditions that could have led to death, but no single CoD was a clear, overburdened healthcare system. While comparing the distribution of IDDs across border regions in France (Grand-Est and Hauts-de-France) and in Belgium (Wallonia), we observed some differences. These may reflect variations in regional policies, the quality of healthcare services, and reporting or coding practices between neighboring border regions. For instance, differences in how health conditions are reported and coded, can significantly influence the way IDDs are recorded in health data. Furthermore, cross-border disparities in health information systems could also contribute to the observed variation.

In the reference year 2017, the ASYRs highlighted significantly higher rates of lung cancer, ischemic heart disease, and stroke among males, while Alzheimer's disease and other dementias were slightly higher among females compared to males. Our results highlighted that overall the distribution for male and female rates, males experience more (premature) deaths compared with females. This finding is consistent with other national burden of disease studies in European countries [15, 43]. For regional variations, the premature mortality patterns among metropolitan regions were similar as in France (overall), whereas these patterns were different in overseas departments. In the metropolitan region, the highest rates were observed for the Haut-de-France region and the lowest for the Occitanie region. The Ile-de-France region, including the capital, held a relatively low position as compared to other metropolitan regions. Whereas in the overseas departments, the highest rates were observed for Mayotte and the lowest for Martinique. Despite some differences in

the ranking, mortality in all the regions was impacted mostly by lung cancer, ischemic heart disease, stroke, and Alzheimer's disease and other dementias. High ASYRs were observed for diabetes mellitus among all overseas departments as compared to metropolitan regions. The 2022 INSEE report underlined a clear contrast in poverty levels between metropolitan regions and overseas departments in France [44]. It highlighted that extreme poverty was not only more prevalent but also significantly more severe in the overseas departments compared to metropolitan France. This disparity is particularly pronounced in the four overseas departments of Guadeloupe, Martinique, Guyane, and Mayotte [44]. Vulnerable groups such as single-parent families, the unemployed and retired individuals are disproportionately affected in these regions. Socioeconomic inequalities in mortality have been documented worldwide and underline the association of health and mortality with socioeconomic status [45–47]. For instance, Saint-Jacques et al., have showed that premature mortality attributable to inequality increased in the years 1995–2005, and about 40% of premature deaths were attributable to deprivation [47]. These results stress the urgent need to address these disparities by improving the living conditions of those most in need, as this is crucial for enhancing overall health outcomes and reducing inequalities across the population.

These estimates may not be comparable with GBD estimates due to different data sources used, differences in redistribution approaches applied and variability in the selection of reference population. Nevertheless, we observed some differences in the proportions of death, such as for Alzheimer's disease and other dementia for 2017 (i.e., GBD estimates: 65% [16–166%], our estimates: 85% [84–86%]). Whereas for the SEYLL estimates, GBD method led to higher premature mortality burden in France for ischemic heart disease (1233 vs. 1007 per 100 000), and stroke (872 vs. 734 per 100 000); and lower estimates for the burden of lung cancer (1396 vs. 1501 per 100 000) and Alzheimer's disease and other dementias (715 vs. 732 per 100 000).

We compared these results with other European countries that conduct their own BoD analyses. For instance, in Belgium, the ranking of ASYRs for lung cancer, ischemic heart disease (IHD) and stroke was the same as in France, the estimates were comparable (lung cancer: 1342 vs 1501, IHD: 1341 vs 1007, stroke: 922 vs 734, respectively) [48]. Whereas for Scotland, IHD ranked first (2585), followed by for lung cancer (1663) and Alzheimer's disease and other dementia (1645), with higher estimates, than those in France and Belgium [49]. In the German BoD study, the YLL rankings were similar, with IHD, lung cancer and stroke in the top positions [43].

This study has some strengths: this is the *first* study that quantifies the French mortality burden across regions using all available multiple causes of death data in France. *Secondly*, we applied the recent method of redistribution developed by Sciensano researchers, which is validated by the BoD experts and published in peer-reviewed journal. We used the GBD reference life table to facilitate the comparison of these results with other studies. Nevertheless, we acknowledge the choice of national or standard GBD life table is of intense debate [35]. There are some limitations of this study, such as the probabilistic nature of the redistribution method requires increased computational capacity. Therefore, to measure the uncertainty, the redistributions were simulated in 30 iterations, providing mean and uncertainty intervals based on these iterations. We took into account only those deaths that occurred in France with residence-based status, which has a marginal impact on the redistribution method. The redistribution method (step 4) may lead to underestimate for categories that are under-reported at the national level. This analysis was performed only on 2017 mortality data.

These key findings implied important implications in public health policy. Our results highlighted that lung cancer, ischemic heart disease, stroke, and Alzheimer's disease and other dementias, are primary contributors to premature mortality. For lung cancer, tobacco smoking is the largest preventable cause and contributes to greater than 80% of the occurrence of this disease [50]. The implementation of comprehensive tobacco control strategies can show significant reductions in lung cancer rates. As indicated by our results, ischemic heart disease and stroke predominantly affect elderly people. Early prevention through lifestyle modifications, such as promoting a healthy diet, regular physical activity, low alcohol consumption and smoking cessation, is crucial. These lifestyle changes can be focused by implementing community-based programs that have proven effective in reducing cardiovascular mortality [51]. With an ageing population, the burden of Alzheimer's disease and other dementias is expected to rise. Public health policies should prioritize early detection [52] and intervention, alongside support for caregivers, to improve the quality of life [53]. Additionally, self-harm and road injuries disproportionately affect younger populations. To reduce the burden of self-harm, public health policies should include mental health education in schools, and accessible mental health services. For example, in Sweden, with the implementation of comprehensive mental health support systems for youth, suicide rate has reduced by 60% over the past 30 years [54]. To minimize road injuries, effective strategies such as stricter enforcement of traffic laws, improved road infrastructure, and educational

campaigns on safe driving should be implemented. The success of Vision Zero initiatives in countries like Sweden and the Netherlands, which aim to eliminate all traffic fatalities, highlights the potential impact of such policies [55]. Above all, many of these harmful risk exposures are patterned according to socio-economic status, and therefore improvements in the fundamental causes of health inequalities would be an effective means to improving population health outcomes.

The regional disparities in premature mortality rates between metropolitan France and overseas departments indicate the need for region-specific strategies. These could include increasing healthcare resources, improving access to medical services, and addressing social determinants of health unique to each region. The overseas departments of France face unique health challenges, necessitating tailored interventions that should consider local cultural and socioeconomic factors. These results would support the tailored strategies specific to age, sex and region to improve the population's health.

Future research should take into account the trend in previous years to analyse the changes in premature mortality over time in France. Moreover, it is necessary to take into account the contribution of risk factors, multimorbidity, health inequalities, make forecasts, and combine these estimates with estimates of Years Lived in Disability (YLD) of diseases, to compute Disability-Adjusted Life Years (DALYs), to further inform preventative and public health interventions to improve population health in France.

Conclusions

This study showed that the four-step probabilistic redistribution method from the BeBoD study was applicable to the French multiple causes of death dataset without significant adaptation. We quantified the full mortality burden for the first time in France at regional level and observed regional differences in the premature mortality burden in 2017, based on a novel probabilistic redistribution method developed by researchers from Sciensano before COVID-19 pandemic. The specific causes of death were assigned to all IDD present in French deaths statistics, by taking into account available multiple causes of death data. The leading causes of death were Alzheimer's disease and other dementias, ischemic heart disease and lung cancer, whereas for premature mortality, lung cancer, ischemic heart disease, stroke and Alzheimer's disease and other dementias. Our results also confirm persisting mortality differences between sexes and French regions. These findings on premature mortality in France provide a valuable foundation for developing comprehensive public health policies. Through

addressing the socio-economic determinants of health, regional disparities, tailoring interventions to specific demographic groups, and focusing on early prevention, policymakers can effectively reduce premature mortality and improve population health outcomes. These estimates are important for future investigations on the contribution of health inequalities and risk factors in France. Future research and monitoring are essential to adapt and refine intervention and prevention strategies, ensuring they meet the evolving health needs of the population.

Abbreviations

YLL	Years of Life Lost
IDD	Ill-defined death
INSERM-CépiDc	Institut National de la Santé et de la Recherche Médicale_ Centre d'épidémiologie sur les causes médicales de Décès de l'Inserm
CoD	Cause of death
GBD	Global Burden of Disease
ICD	International statistical classification of diseases and related health problems
WHO	World Health Organisation
BeBOD	Belgium Burden of Disease
SEYLL	Standard Expected Years of Life Lost
DALYs	Disability-Adjusted Life Years
YLD	Years of life lived with disability
SNDS	Système National de Données Santé
INSEE	Institute national de la statistique et des études économiques
NCD	Non-communicable diseases
MCOD	Multiple Causes of Death
ASYRS	Age-standardized SEYLL rates

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-025-23559-6>.

Additional file 1. It describes the ill-defined deaths (IDDs) total, with and without R-codes in France in 2017.

Additional file 2. It describes the IDDs by age group, sex, regions and place of death in France in 2017.

Additional file 3. It describes the Age-standardized mortality rates by regions—GBD level 1 in France in 2017.

Additional file 4. It describes the Age-standardized mortality rates of the top 10 causes by regions—GBD level 2 in France in 2017.

Additional file 5. It describes the supplementary figures S1 – S6.

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Authors' contributions

RH and BD generated the idea for the study. RH, AS and BD adopted the methodological approach to French context. AS and BD designed the analyses. RH and PC supported data curation. RH and BD carried out all the analyses. RH performed the visualization of all results and NM created Figs 1 and 6. RH drafted the original manuscript. All other authors provided critical inputs

into the method, interpretation of results and revisions to the manuscript. All authors reviewed the manuscript and approved the final draft.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request. The underlying micro data that support the findings of this study are available from Santé Publique France but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publically available.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the ethical standards of the Helsinki Declaration of 1975 (revised in 2013).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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